

# DBscreamZ

Pitch-tracking formant scream  
synthesiser

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User manual  
& preset field guide

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# Contents

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|    |                    |    |
|----|--------------------|----|
| 1  | What this is       | 3  |
| 2  | First sound        | 5  |
| 3  | The controls       | 6  |
| 4  | The three menus    | 9  |
| 5  | Memory             | 12 |
| 6  | Charge mode        | 14 |
| 7  | Reading a card     | 17 |
| 8  | The eight voices   | 20 |
|    | Wukong             | 20 |
|    | Prince             | 22 |
|    | Rice               | 24 |
|    | Piccolo            | 26 |
|    | Boo                | 28 |
|    | Fling              | 30 |
|    | Ki-Ki              | 32 |
|    | Master             | 34 |
| 9  | Every gesture      | 36 |
| 10 | When it misbehaves | 38 |
| 11 | Specifications     | 40 |

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# 1 What this is

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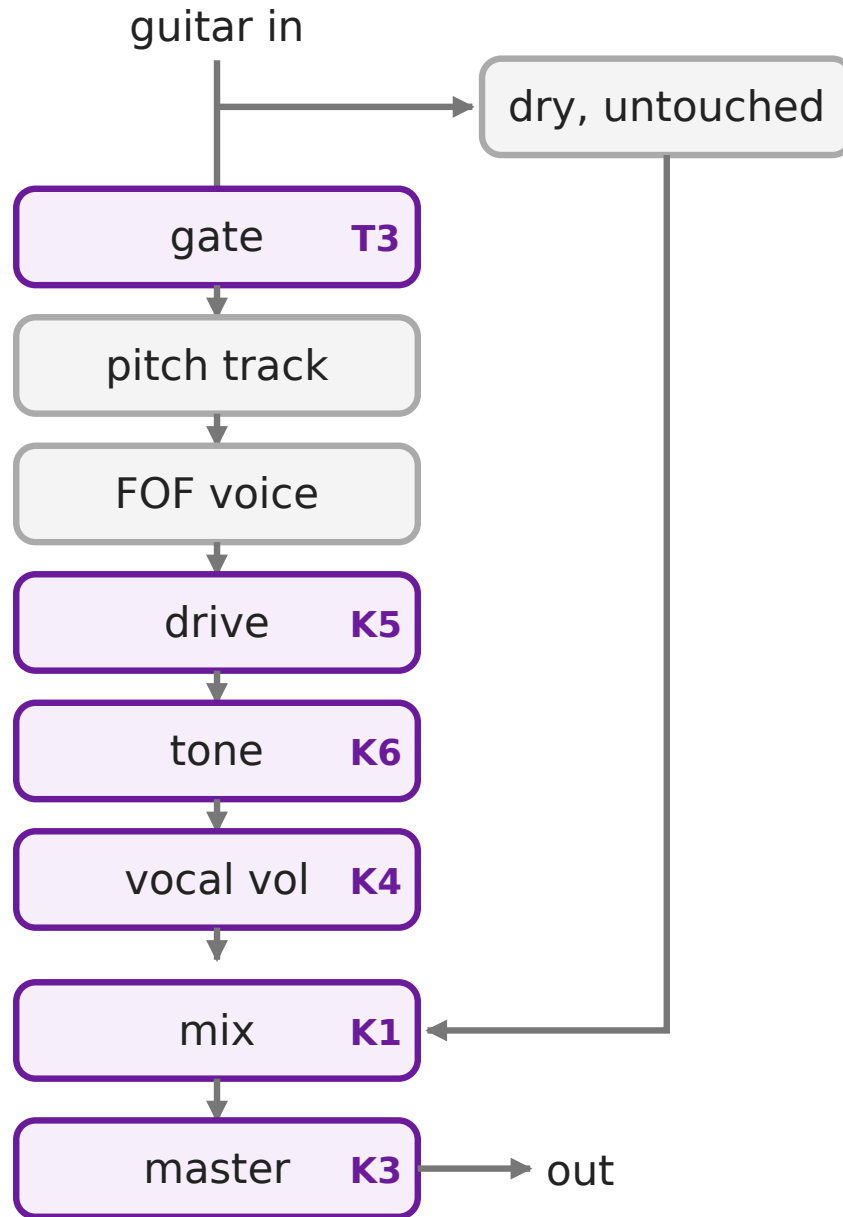
Play a note. A synthesised voice screams it back at you, tracking your pitch as you bend, slide and vibrato.

The voice is not a sample and not a filter draped over your guitar. It is built from scratch every block: a pitch tracker follows your fundamental, and a bank of formant grains rebuilds a vowel at that pitch. Three formants (F1, F2, F3) decide which vowel and how bright it is. There is no noise anywhere in the voice path, which is why it screams rather than hisses.

Your dry signal is never processed. It is passed through untouched and crossfaded against the voice at the end, so Mix at 7:00 is your guitar, exactly as it went in.

## Signal flow

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The gate watches your input envelope and decides when the voice may sound. It never gates your dry signal.

## 2 First sound

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1. Guitar into **IN**, amp into **OUT**, 9 V DC into the barrel jack. The pedal boots **bypassed**, both LEDs off.
2. Middle toggle **up** (Set 1). Put the other two toggles in the **middle**.
3. Tap the **RIGHT** footswitch. The right LED goes solid: Wukong is engaged.
4. Play single notes, cleanly, one at a time. The voice follows your pitch.
5. Nothing screaming? Turn **knob 1 (Mix)** clockwise and set the **right toggle to the middle**.
6. Now stomp **both** footswitches at once and hold. That is Charge mode.

### ONE NOTE AT A TIME

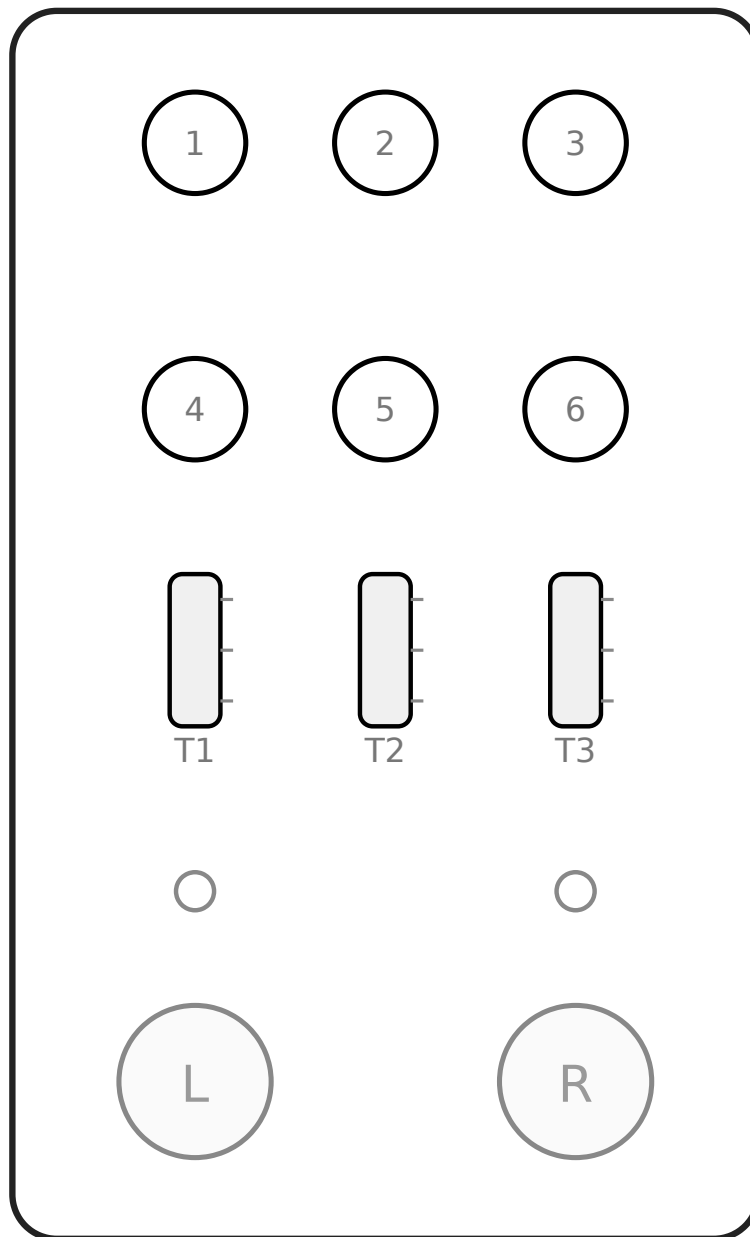
Pitch tracking needs a single fundamental. Chords, open strings ringing under a lead line, and heavy pick noise all confuse it. Mute what you are not playing.

Tap the same footswitch again to bypass. Tap the **LEFT** footswitch instead for the other voice on this page (Prince). Flip the middle toggle **down** for Set 2: Rice on the right, Piccolo on the left.

Everything you change is live immediately and lost at power off until you save it. Saving is in section 5.

### 3 The controls

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Six knobs in two rows of three, numbered left to right: **1 2 3** on top, **4 5 6** below. Three toggles, each up, middle or down. Two footswitches, an LED [light-emitting diode] above each.

## Knob travel

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Knobs run **7:00** fully anticlockwise, through **12:00** at centre, to **5:00** fully clockwise. Every position in this booklet is a clock face.

## What the knobs do

---

|           | Menu 1     | Menu 2       | Menu 3       |
|-----------|------------|--------------|--------------|
| <b>K1</b> | Mix        | F1           | F3           |
| <b>K2</b> | Glide      | F1 bandwidth | F3 bandwidth |
| <b>K3</b> | Master vol | F1 amount    | F3 amount    |
| <b>K4</b> | Vocal vol  | F2           | Voices       |
| <b>K5</b> | Drive      | F2 bandwidth | Detune       |
| <b>K6</b> | Tone       | F2 amount    | Aspiration   |

Menu 3: **K4** steps through the eight voice counts in eight equal bands, so it always lands on a whole number. Turn **K6** fully anticlockwise and the aspiration switches **off**: the last sliver before 7:00 reads as a hard zero, so there is no breath in the voice at all rather than a residual trickle.

## What the toggles do

---

|                     | Up    | Middle   | Down  |
|---------------------|-------|----------|-------|
| <b>T1</b><br>Octave | +1    | 0        | -1    |
| <b>T2</b><br>Page   | Set 1 | Freeform | Set 2 |
| <b>T3</b><br>Gate   | high  | medium   | low   |

---

That is with no menu latched. While a menu *is* latched the toggles configure Charge mode instead, and nothing else. See section 6.

The gate decides how loud you must play before the voice speaks. High needs a firm attack and stays quiet between notes; low lets quiet playing through and hangs on longer.



## 4 The three menus

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Six knobs, eighteen parameters. What a knob does depends on which menu is latched.

| Menu | How to get there |
|------|------------------|
|------|------------------|

---

|     |                                    |
|-----|------------------------------------|
| ● 1 | Where you are with nothing latched |
|-----|------------------------------------|

---

|     |                                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ● 2 | Hold the <b>RIGHT</b> footswitch. About 1 second in it latches under your foot: right LED blinks, left goes dark. Letting go changes nothing. |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|

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|     |                                                                                 |
|-----|---------------------------------------------------------------------------------|
| ● 3 | Hold the <b>LEFT</b> footswitch the same way. Left LED blinks, right goes dark. |
|-----|---------------------------------------------------------------------------------|

---

A menu **latches**: it stays until you leave it.

- Tap the **other** footswitch to jump straight to the other menu.
- Tap the **blinking side's own** footswitch to drop back to menu 1.

### **KNOB PICKUP: NOTHING JUMPS**

Every time you change menu, or recall a voice, all six knobs go **inert**. A knob does nothing until you **move it**, about a fingernail's nudge. The moment it moves it takes over its parameter, from wherever it is physically sitting.

A knob you do not touch cannot hurt the sound. A knob you do touch grabs control at once: it does not wait for you to sweep back to the stored value.

## What the formant controls mean

---

**F1 and F2** decide which vowel the voice is shouting. Move them and the scream changes from an *ah* to an *ee* to an *oh*.

**F3** mostly decides how bright and nasal it sounds.

**Bandwidth** is how wide each resonance is: narrow is more vocal, wide is more washed out. **Amount** is how much of that formant you get.

**Voices** is how many copies of the scream sing at once, one to eight, and **Detune** is how far apart they are tuned, in cents. Together they are the thickness of the voice: one voice is bare and focused, eight spread wide is a crowd of one person.

**Aspiration** is breath. It is not noise: there is no hiss generator anywhere in this pedal. It is two extra inharmonic tones inside the voice itself, which is why it tears rather than hisses.

## 5 Memory

---

The middle toggle picks a **page**. Each page has two **slots**, one per footswitch. Four stored voices.

| T2   | Page     | Left                                                                                 | Right  |
|------|----------|--------------------------------------------------------------------------------------|--------|
| Up   | Set 1    | Prince                                                                               | Wukong |
| Mid  | Freeform | no slots: the footswitches engage and bypass whatever the knobs are set to right now |        |
| Down | Set 2    | Piccolo                                                                              | Rice   |

Those four are what it ships with. Save over any of them: the recipes to dial them back are in section 8.

## Saving

---

Everything you tweak is live but volatile. To save:

1. Hold **one** footswitch past 1 second.
2. **While still holding it**, press the other.

The **held** side picks the slot. Hold RIGHT and press LEFT and you have saved to the **right** slot of the current page. Both LEDs blink three times.

### **FREEFORM CANNOT SAVE**

It has no slot, so the save is refused and both LEDs give one short double-flicker. Pick Set 1 or Set 2 first.

Leave any latched menu before saving. Menus never save, and your edits survive the exit.

The held footswitch latches its own menu on the way past 1 second, the same as any hold. The second press drops that menu again and saves, so you finish where you began.

## 6 Charge mode

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With a voice engaged and **no menu latched**, stomp **both** footswitches together and hold. The scream charges: gain swells, pitch sweeps, breath tears into the voice, and the LEDs alternate faster and faster as it builds. Release and it winds down.

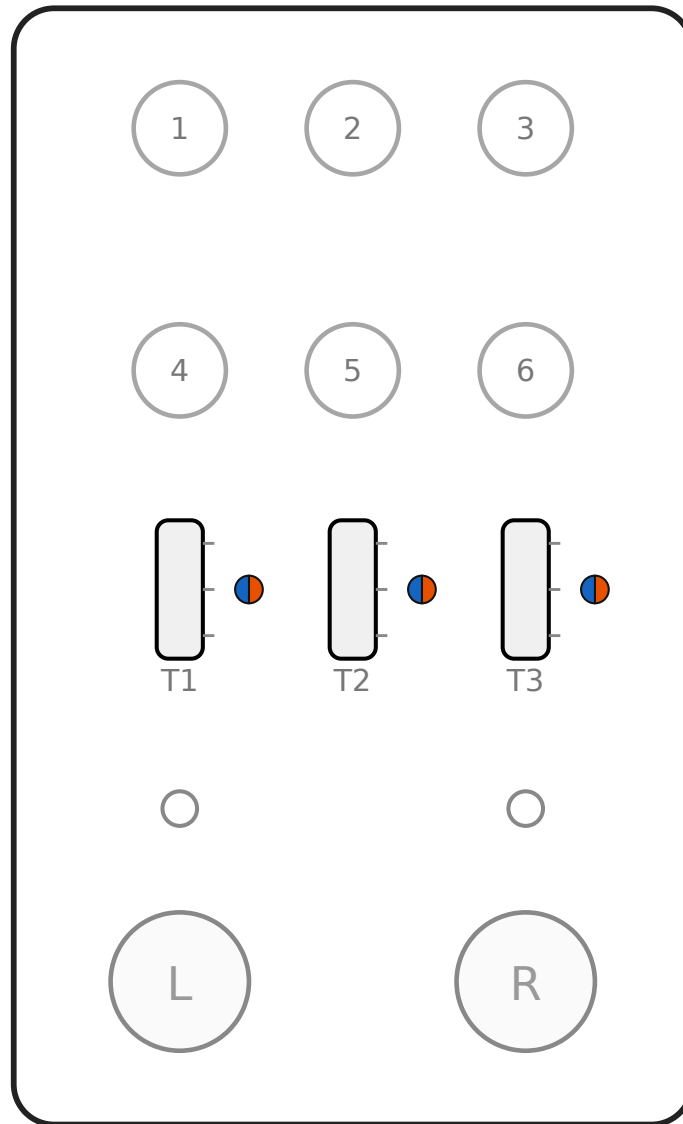
It lasts exactly as long as you hold it. No latch, no timeout.

**Engaged is the whole condition.** The same two-stomp hold while *bypassed* is the firmware-update gesture instead, so charge can never reach it and it can never reach charge. See section 11.

Charge is configured on the **toggles**, while a menu is latched. Latch menu 2 (hold RIGHT) for the orange settings, menu 3 (hold LEFT) for the blue ones. The knobs do nothing here.

## Factory charge settings

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Every toggle centred. Each dot is half orange and half blue because both menus want that switch in the middle: orange on the right, blue on the left.

This configuration is **global**: one setting shared by every voice, kept across power cycles, and never touched by saving or recalling. That is why it is not on the character cards.

## ● With menu 2 latched

|                             | Up                  | Middle               | Down             |
|-----------------------------|---------------------|----------------------|------------------|
| <b>T1</b><br>Gain           | Above 9000!         | on                   | off              |
| <b>T2</b><br>Charge<br>time | Birit Spomb<br>~6 s | Hamekameka<br>~2.5 s | punch<br>~0.75 s |
| <b>T3</b><br>Decay          | fast                | slow                 | off (instant)    |

## ● With menu 3 latched

|                         | Up       | Middle | Down |
|-------------------------|----------|--------|------|
| <b>T1</b><br>Pitch      | rise     | fall   | off  |
| <b>T2</b><br>Tone       | brighter | darker | off  |
| <b>T3</b><br>Aspiration | high     | low    | off  |

Centred means the charge pitch *falls*. For the rising power-up, latch menu 3 and flick T1 up.



## 7 Reading a card

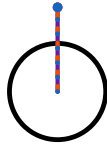
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Each card is the pedal face with a pointer drawn out of every knob, one per menu, ending at the position that menu wants. **Twenty-one positions:** six knobs in each of three menus, plus three toggles. Set every one. None can be assumed: the knobs are wherever you left them.

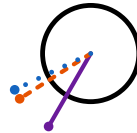
-  **Menu 1**, solid pointer  
default layer
-  **Menu 2**, dashed pointer  
hold RIGHT
-  **Menu 3**, dotted pointer  
hold LEFT

Told apart by colour *and* by line, so a photocopy still works. Menus 2 and 3 wear the colour of the LED on the side you hold to reach them: orange for the right, blue for the left.

Every pointer sits at its exact position, including when menus agree. The line styles interleave, so where two or three menus want the same knob in the same place you get one pointer striped in their colours rather than three lines fighting over the same pixels.



**All three agree.**  
Knob 3 on every  
card: 12:00 in all  
three menus.



**One apart, two  
close.** Knob 2 on  
every card.



**A tight cluster.**  
Fling's knob 4,  
inside 14 minutes.

### THE TABLE IS THE EXACT WORD

Pointers are drawn true, but a knob is a blunt instrument and two pointers a few minutes apart look like one. Each card is followed by the same twenty-one positions as a table. When the drawing is ambiguous, read the table.

## Toggles on a card

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Toggle dots sit to the right of the switch, level with the position they want.

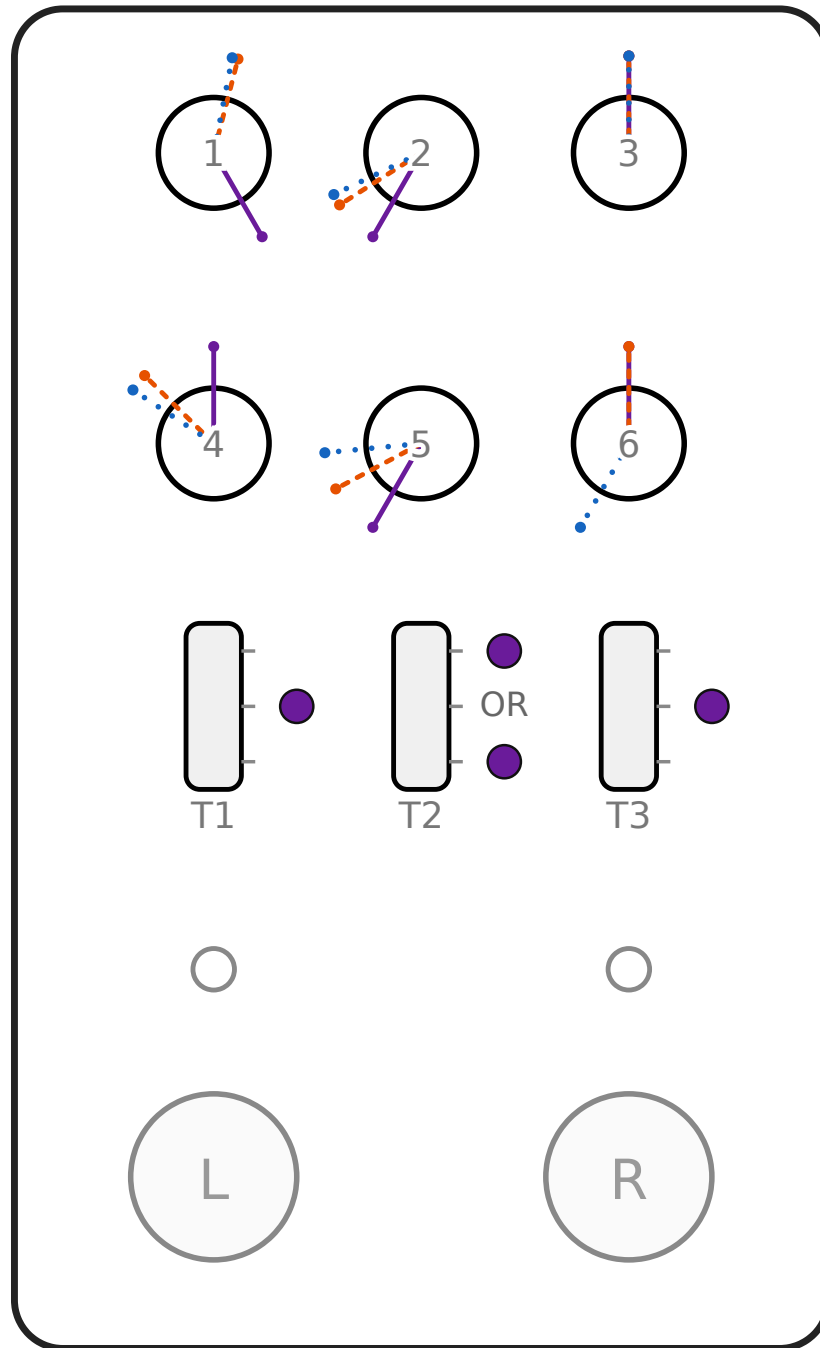
The middle toggle always shows **two** dots, marked OR. A voice can live on either Set, and which one you pick is up to you. It cannot live on Freeform, because Freeform has no slot to save into.

The other two toggles show one dot each. Every character wants octave 0 and the medium gate; both are the middle position.

Four of the eight have a factory home. The other four are recipes: dial them in and save them wherever you like.

# Wukong *The default shout*

Factory: Set 1, RIGHT slot



Mid-placed formants over the reference three-voice stack, with no breath at all. The straightest read of the effect: loud, open, clean. Start here.

## Wukong: every position

---

### ● Menu 1 (default)

|    |            |            |             |
|----|------------|------------|-------------|
| K1 | Mix        | full voice | 100% · 5:00 |
| K2 | Glide      | 0 ms       | 0% · 7:00   |
| K3 | Master vol | unity      | 50% · 12:00 |
| K4 | Vocal vol  | unity      | 50% · 12:00 |
| K5 | Drive      | clean      | 0% · 7:00   |
| K6 | Tone       | flat       | 50% · 12:00 |

---

### ● Menu 2 (hold RIGHT)

|    |              |          |             |
|----|--------------|----------|-------------|
| K1 | F1           | 858.4 Hz | 55% · 12:29 |
| K2 | F1 bandwidth | 32.5 Hz  | 9% · 7:55   |
| K3 | F1 amount    | 1.0      | 50% · 12:00 |
| K4 | F2           | 1234 Hz  | 35% · 10:29 |
| K5 | F2 bandwidth | 47.5 Hz  | 11% · 8:04  |
| K6 | F2 amount    | 1.0      | 50% · 12:00 |

---

### ● Menu 3 (hold LEFT)

|    |              |           |             |
|----|--------------|-----------|-------------|
| K1 | F3           | 3111.7 Hz | 54% · 12:22 |
| K2 | F3 bandwidth | 62.5 Hz   | 12% · 8:09  |
| K3 | F3 amount    | 1.0       | 50% · 12:00 |
| K4 | Voices       | 3         | 31% · 10:07 |
| K5 | Detune       | 11 ct     | 18% · 8:49  |
| K6 | Aspiration   | 0         | 0% · 7:00   |

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### ● Toggles

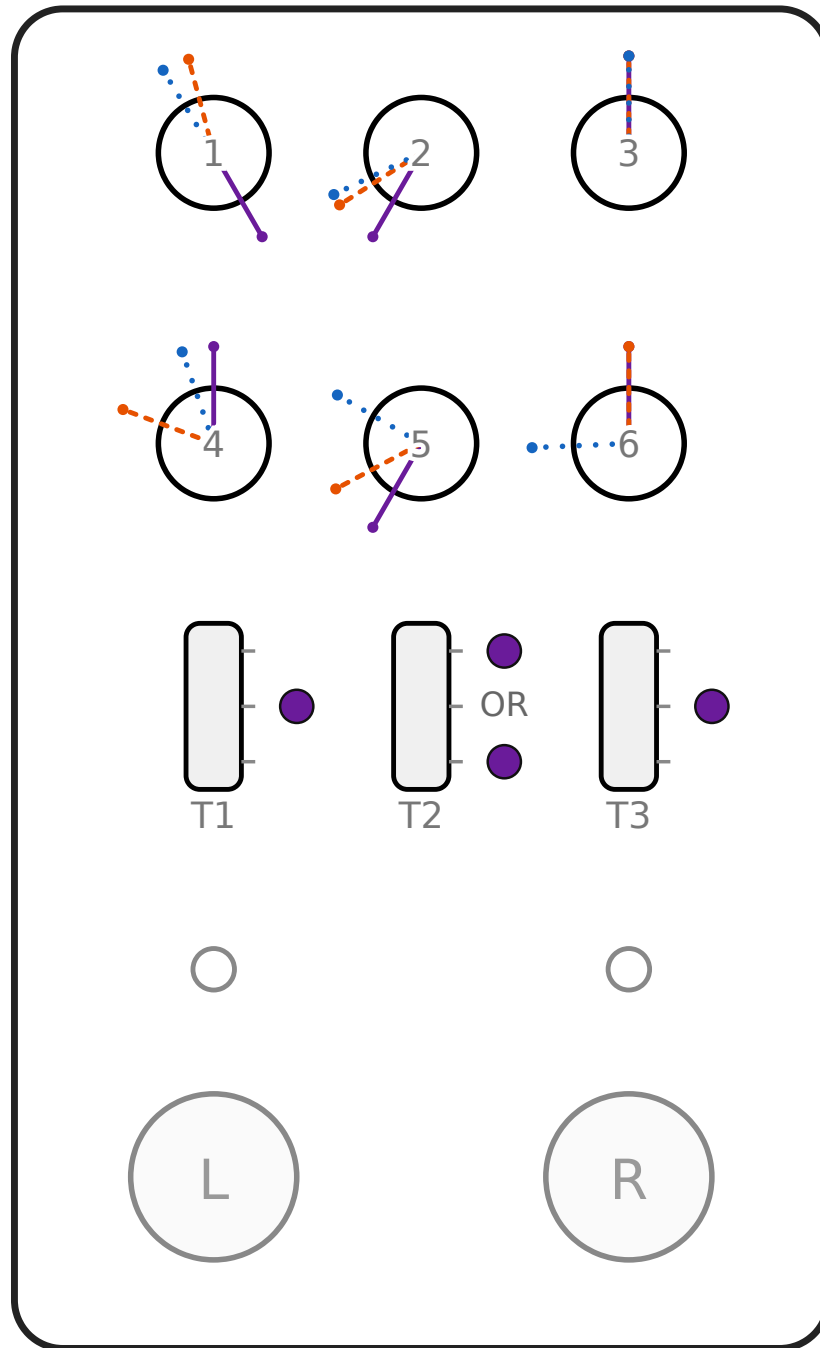
|    |             |                |            |
|----|-------------|----------------|------------|
| T1 | Octave      | 0              | Middle     |
| T2 | Memory page | Set 1 or Set 2 | Up or Down |
| T3 | Gate        | medium         | Middle     |

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# Prince *Clenched teeth*

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Factory: Set 1, LEFT slot



Formants well below Wukong under a wider four-voice stack and a little breath, so it lands darker and rougher. Reads as effort rather than power.

## Prince: every position

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### ● Menu 1 (default)

|    |            |            |             |
|----|------------|------------|-------------|
| K1 | Mix        | full voice | 100% · 5:00 |
| K2 | Glide      | 0 ms       | 0% · 7:00   |
| K3 | Master vol | unity      | 50% · 12:00 |
| K4 | Vocal vol  | unity      | 50% · 12:00 |
| K5 | Drive      | clean      | 0% · 7:00   |
| K6 | Tone       | flat       | 50% · 12:00 |

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### ● Menu 2 (hold RIGHT)

|    |              |          |             |
|----|--------------|----------|-------------|
| K1 | F1           | 741.6 Hz | 45% · 11:30 |
| K2 | F1 bandwidth | 32.5 Hz  | 9% · 7:55   |
| K3 | F1 amount    | 1.0      | 50% · 12:00 |
| K4 | F2           | 1066 Hz  | 27% · 9:41  |
| K5 | F2 bandwidth | 47.5 Hz  | 11% · 8:04  |
| K6 | F2 amount    | 1.0      | 50% · 12:00 |

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### ● Menu 3 (hold LEFT)

|    |              |           |             |
|----|--------------|-----------|-------------|
| K1 | F3           | 2688.3 Hz | 40% · 10:57 |
| K2 | F3 bandwidth | 62.5 Hz   | 12% · 8:09  |
| K3 | F3 amount    | 1.0       | 50% · 12:00 |
| K4 | Voices       | 4         | 44% · 11:22 |
| K5 | Detune       | 18 ct     | 30% · 10:00 |
| K6 | Aspiration   | 0.15      | 19% · 8:55  |

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### ● Toggles

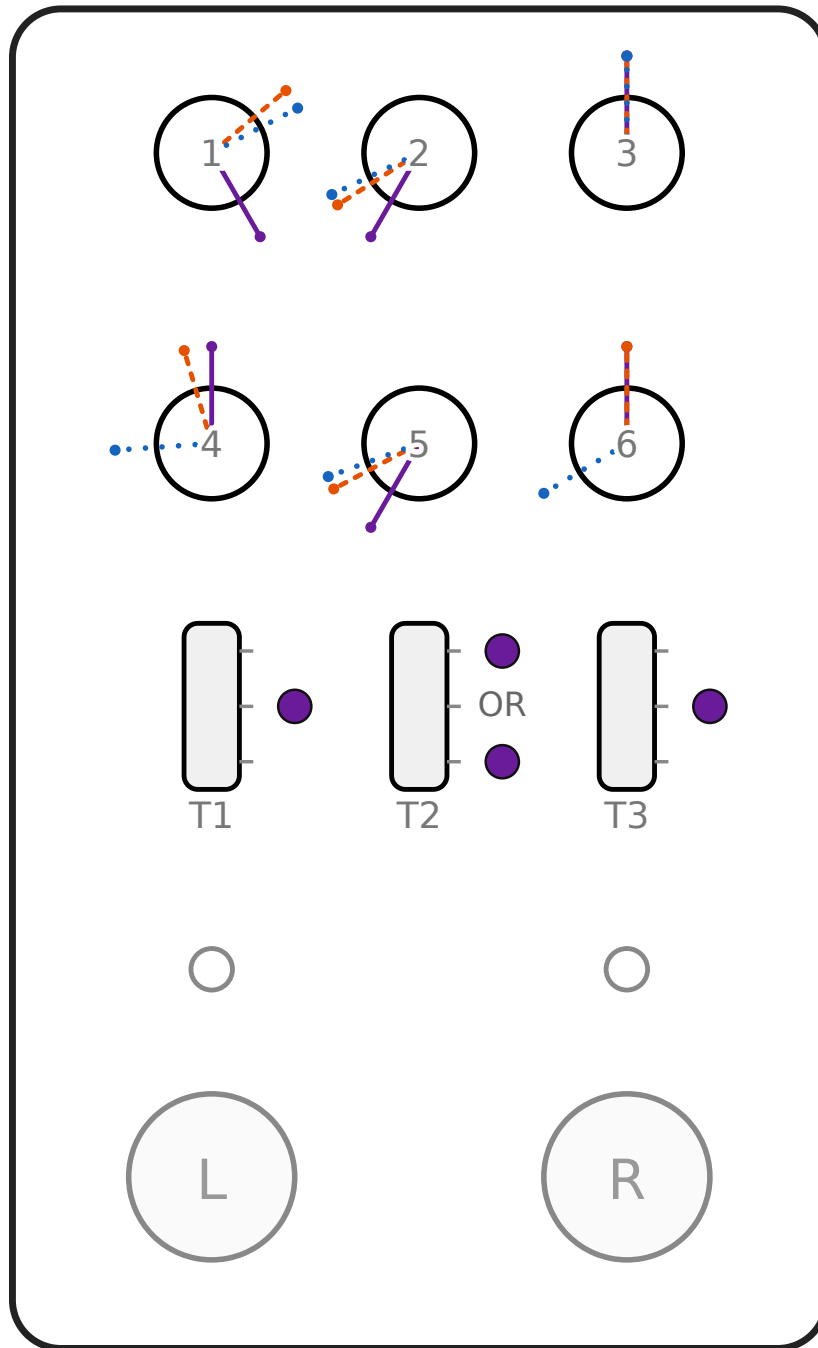
|    |             |                |            |
|----|-------------|----------------|------------|
| T1 | Octave      | 0              | Middle     |
| T2 | Memory page | Set 1 or Set 2 | Up or Down |
| T3 | Gate        | medium         | Middle     |

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# Rice *Clean and glassy*

---

Factory: Set 2, RIGHT slot



The highest F3 of the eight over the thinnest stack in the factory set: two voices, barely detuned. Bright and hard-edged. Cuts without sounding strained.



## Rice: every position

### ● Menu 1 (default)

|    |            |            |             |
|----|------------|------------|-------------|
| K1 | Mix        | full voice | 100% · 5:00 |
| K2 | Glide      | 0 ms       | 0% · 7:00   |
| K3 | Master vol | unity      | 50% · 12:00 |
| K4 | Vocal vol  | unity      | 50% · 12:00 |
| K5 | Drive      | clean      | 0% · 7:00   |
| K6 | Tone       | flat       | 50% · 12:00 |

### ● Menu 2 (hold RIGHT)

|    |              |           |             |
|----|--------------|-----------|-------------|
| K1 | F1           | 1000 Hz   | 67% · 1:40  |
| K2 | F1 bandwidth | 32.5 Hz   | 9% · 7:55   |
| K3 | F1 amount    | 1.0       | 50% · 12:00 |
| K4 | F2           | 1437.5 Hz | 45% · 11:27 |
| K5 | F2 bandwidth | 47.5 Hz   | 11% · 8:04  |
| K6 | F2 amount    | 1.0       | 50% · 12:00 |

### ● Menu 3 (hold LEFT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F3           | 3625 Hz | 71% · 2:05  |
| K2 | F3 bandwidth | 62.5 Hz | 12% · 8:09  |
| K3 | F3 amount    | 1.0     | 50% · 12:00 |
| K4 | Voices       | 2       | 19% · 8:52  |
| K5 | Detune       | 8 ct    | 13% · 8:20  |
| K6 | Aspiration   | 0.05    | 10% · 7:58  |

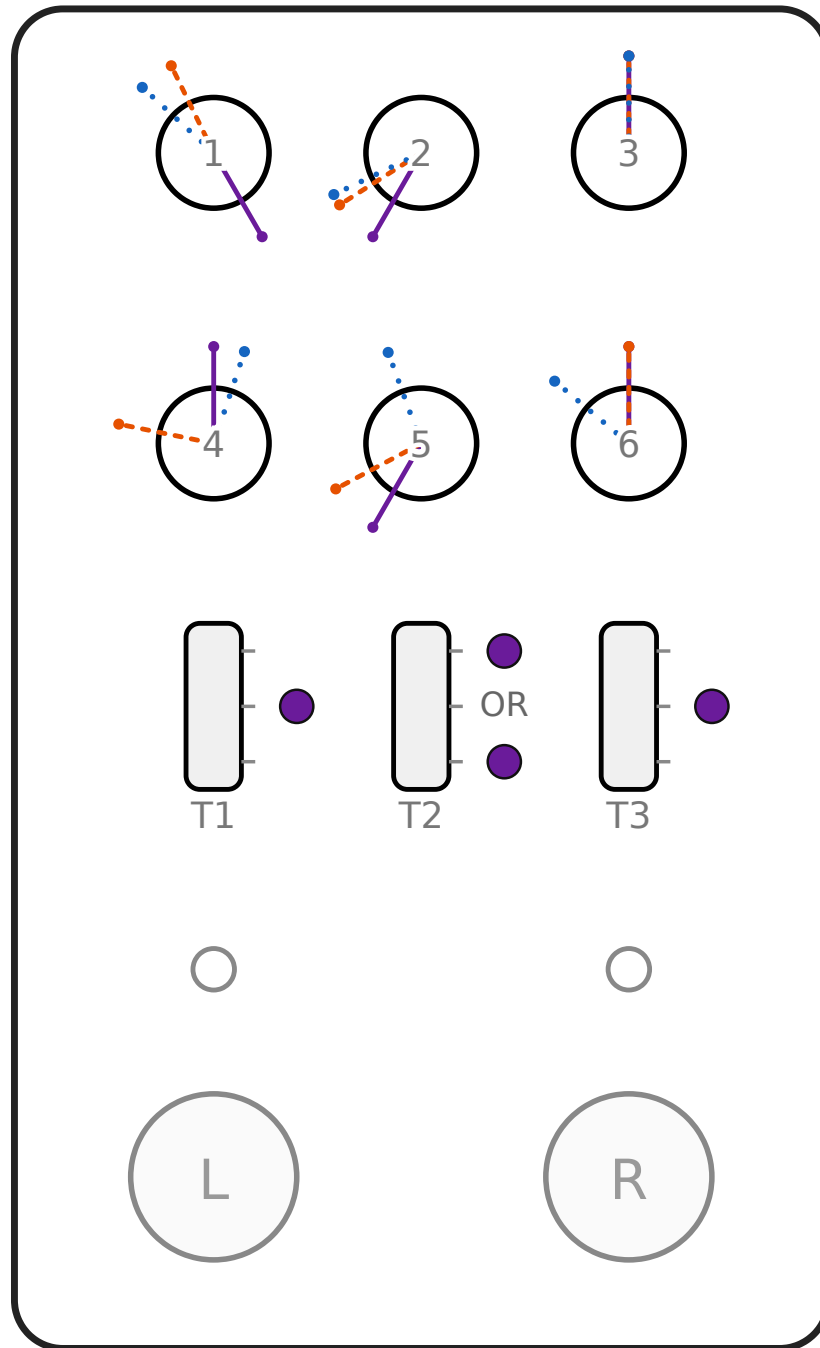
### ● Toggles

|    |             |                |            |
|----|-------------|----------------|------------|
| T1 | Octave      | 0              | Middle     |
| T2 | Memory page | Set 1 or Set 2 | Up or Down |
| T3 | Gate        | medium         | Middle     |

# Piccolo *Low growl*

---

Factory: Set 2, LEFT slot



The lowest formants in the factory set under the thickest factory stack, five voices spread wide. Thick and throaty rather than piercing.

## Piccolo: every position

---

### ● Menu 1 (default)

|    |            |            |             |
|----|------------|------------|-------------|
| K1 | Mix        | full voice | 100% · 5:00 |
| K2 | Glide      | 0 ms       | 0% · 7:00   |
| K3 | Master vol | unity      | 50% · 12:00 |
| K4 | Vocal vol  | unity      | 50% · 12:00 |
| K5 | Drive      | clean      | 0% · 7:00   |
| K6 | Tone       | flat       | 50% · 12:00 |

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### ● Menu 2 (hold RIGHT)

|    |              |           |             |
|----|--------------|-----------|-------------|
| K1 | F1           | 697.6 Hz  | 41% · 11:08 |
| K2 | F1 bandwidth | 32.5 Hz   | 9% · 7:55   |
| K3 | F1 amount    | 1.0       | 50% · 12:00 |
| K4 | F2           | 1002.8 Hz | 24% · 9:23  |
| K5 | F2 bandwidth | 47.5 Hz   | 11% · 8:04  |
| K6 | F2 amount    | 1.0       | 50% · 12:00 |

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### ● Menu 3 (hold LEFT)

|    |              |           |             |
|----|--------------|-----------|-------------|
| K1 | F3           | 2528.8 Hz | 34% · 10:25 |
| K2 | F3 bandwidth | 62.5 Hz   | 12% · 8:09  |
| K3 | F3 amount    | 1.0       | 50% · 12:00 |
| K4 | Voices       | 5         | 56% · 12:37 |
| K5 | Detune       | 26 ct     | 43% · 11:20 |
| K6 | Aspiration   | 0.3       | 34% · 10:20 |

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### ● Toggles

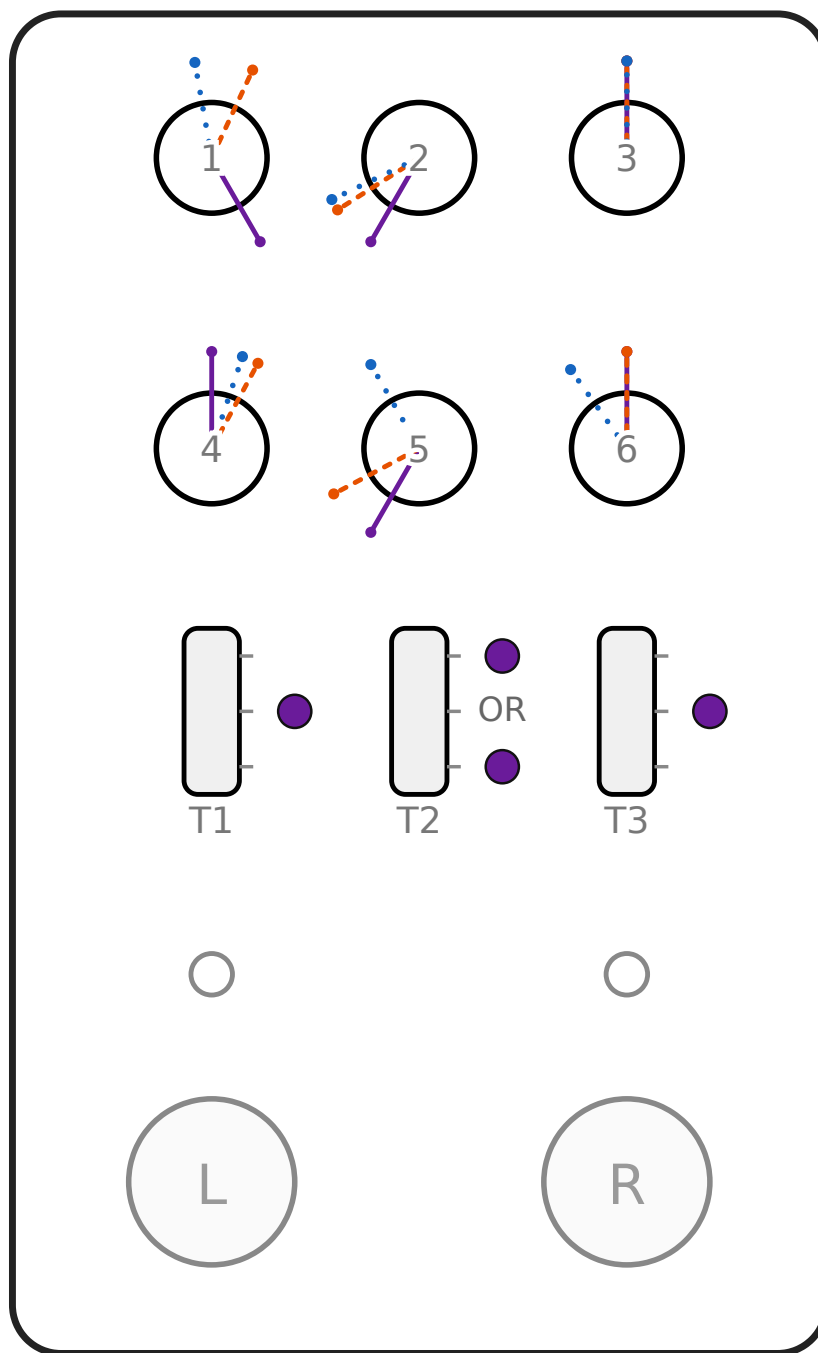
|    |             |                |            |
|----|-------------|----------------|------------|
| T1 | Octave      | 0              | Middle     |
| T2 | Memory page | Set 1 or Set 2 | Up or Down |
| T3 | Gate        | medium         | Middle     |

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## Boo *Rubbery and hollow*

---

No factory slot: save it to any page



A wide F1-to-F2 gap over a low F3, thickened by five detuned voices and the second-most breath here. A big soft body rather than an edge.

## Boo: every position

---

### ● Menu 1 (default)

|    |            |            |             |
|----|------------|------------|-------------|
| K1 | Mix        | full voice | 100% · 5:00 |
| K2 | Glide      | 0 ms       | 0% · 7:00   |
| K3 | Master vol | unity      | 50% · 12:00 |
| K4 | Vocal vol  | unity      | 50% · 12:00 |
| K5 | Drive      | clean      | 0% · 7:00   |
| K6 | Tone       | flat       | 50% · 12:00 |

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### ● Menu 2 (hold RIGHT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F1           | 900 Hz  | 58% · 12:50 |
| K2 | F1 bandwidth | 32.5 Hz | 9% · 7:55   |
| K3 | F1 amount    | 1.0     | 50% · 12:00 |
| K4 | F2           | 1750 Hz | 60% · 12:57 |
| K5 | F2 bandwidth | 47.5 Hz | 11% · 8:04  |
| K6 | F2 amount    | 1.0     | 50% · 12:00 |

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### ● Menu 3 (hold LEFT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F3           | 2900 Hz | 47% · 11:40 |
| K2 | F3 bandwidth | 62.5 Hz | 12% · 8:09  |
| K3 | F3 amount    | 1.0     | 50% · 12:00 |
| K4 | Voices       | 5       | 56% · 12:37 |
| K5 | Detune       | 24 ct   | 40% · 11:00 |
| K6 | Aspiration   | 0.35    | 38% · 10:49 |

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### ● Toggles

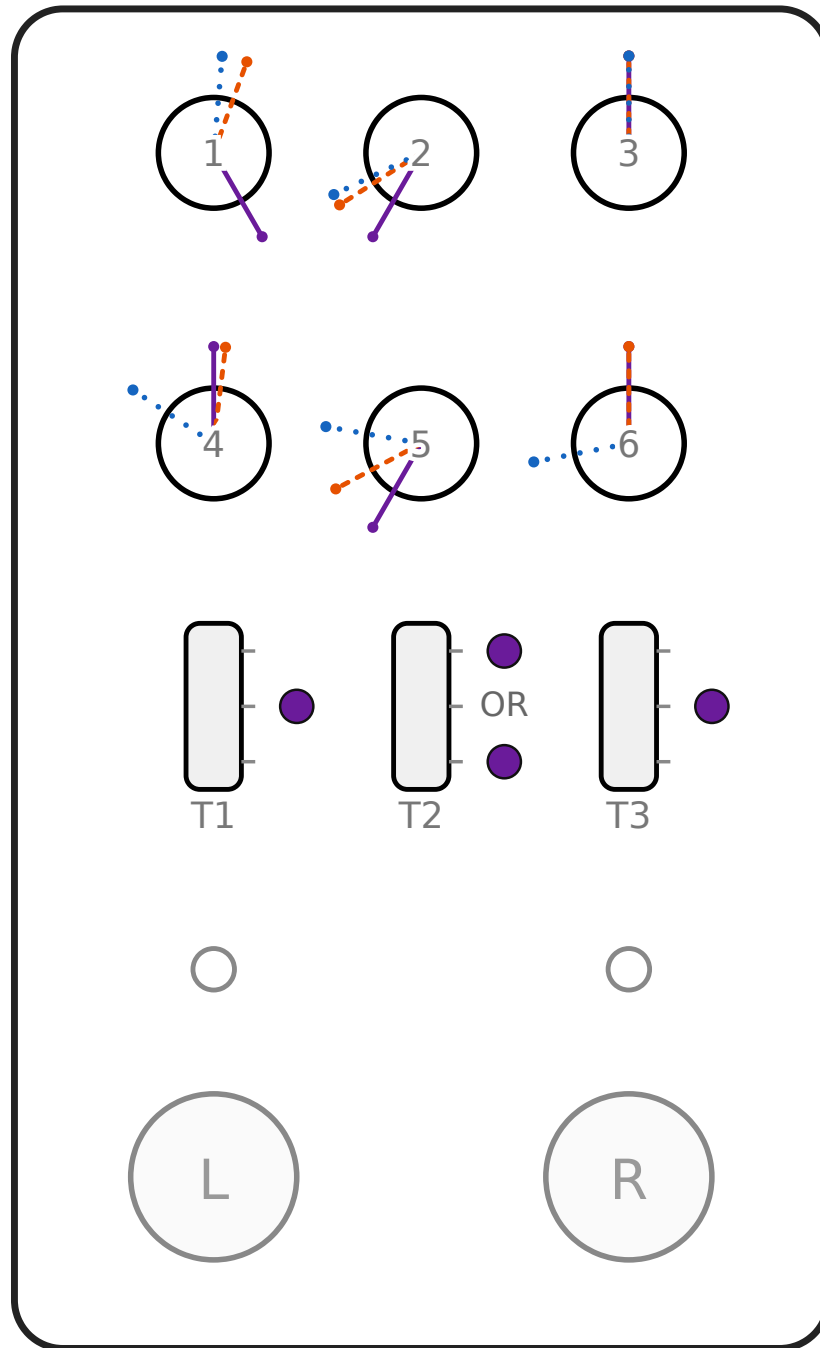
|    |             |                |            |
|----|-------------|----------------|------------|
| T1 | Octave      | 0              | Middle     |
| T2 | Memory page | Set 1 or Set 2 | Up or Down |
| T3 | Gate        | medium         | Middle     |

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# Fling *Light and quick*

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No factory slot: save it to any page



Between Wukong and Rice on every formant, with a modest three-voice stack so the edge stays on the formants. The most usable of the eight under a band.

## Fling: every position

---

### ● Menu 1 (default)

|    |            |            |             |
|----|------------|------------|-------------|
| K1 | Mix        | full voice | 100% · 5:00 |
| K2 | Glide      | 0 ms       | 0% · 7:00   |
| K3 | Master vol | unity      | 50% · 12:00 |
| K4 | Vocal vol  | unity      | 50% · 12:00 |
| K5 | Drive      | clean      | 0% · 7:00   |
| K6 | Tone       | flat       | 50% · 12:00 |

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### ● Menu 2 (hold RIGHT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F1           | 880 Hz  | 57% · 12:40 |
| K2 | F1 bandwidth | 32.5 Hz | 9% · 7:55   |
| K3 | F1 amount    | 1.0     | 50% · 12:00 |
| K4 | F2           | 1600 Hz | 52% · 12:14 |
| K5 | F2 bandwidth | 47.5 Hz | 11% · 8:04  |
| K6 | F2 amount    | 1.0     | 50% · 12:00 |

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### ● Menu 3 (hold LEFT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F3           | 3050 Hz | 52% · 12:10 |
| K2 | F3 bandwidth | 62.5 Hz | 12% · 8:09  |
| K3 | F3 amount    | 1.0     | 50% · 12:00 |
| K4 | Voices       | 3       | 31% · 10:07 |
| K5 | Detune       | 14 ct   | 23% · 9:20  |
| K6 | Aspiration   | 0.12    | 16% · 8:38  |

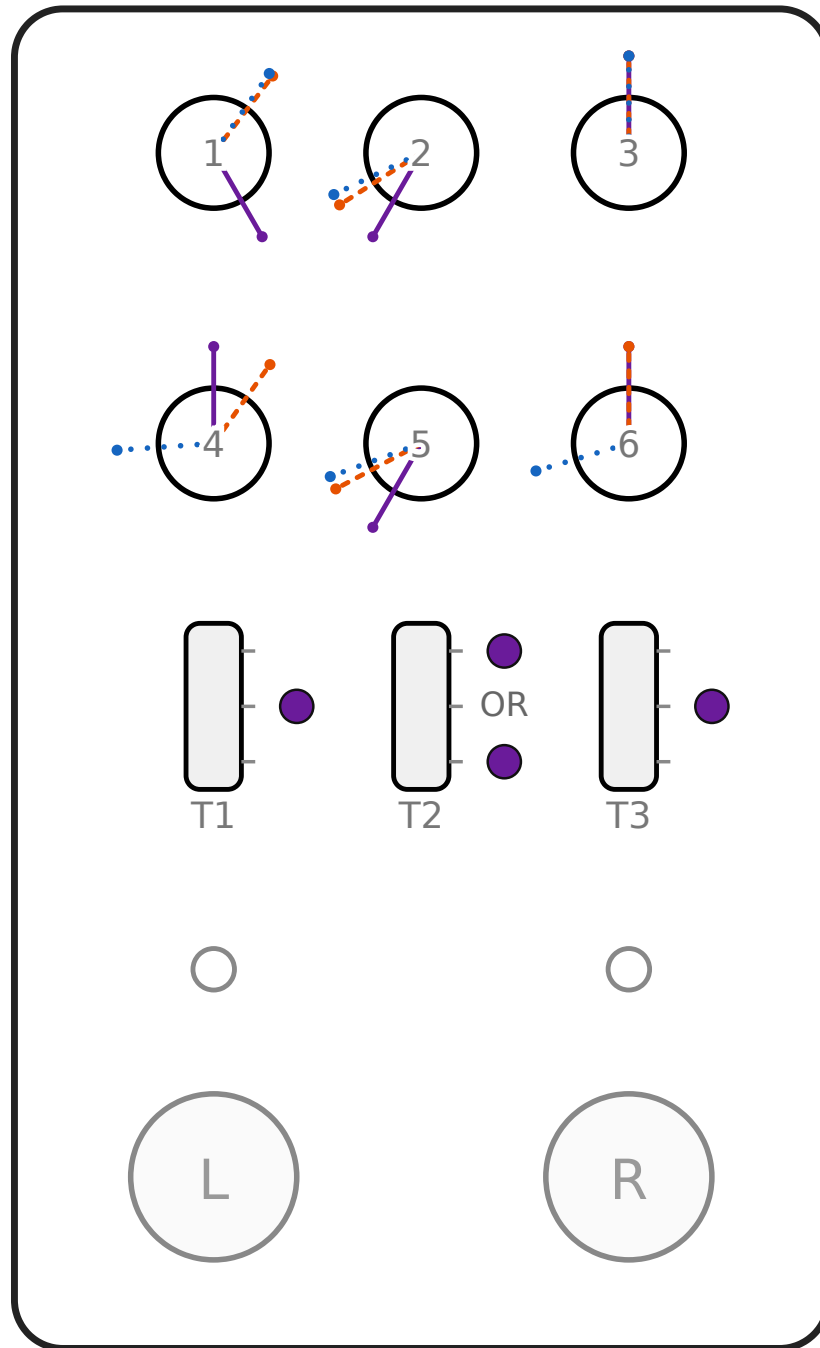
---

### ● Toggles

|    |             |                |            |
|----|-------------|----------------|------------|
| T1 | Octave      | 0              | Middle     |
| T2 | Memory page | Set 1 or Set 2 | Up or Down |
| T3 | Gate        | medium         | Middle     |

---

No factory slot: save it to any page



The highest F2 of the eight and the thinnest stack of the eight, all three formants crowded high with nothing blunting them. Shrill, nasal, annoyed.



## Ki-Ki: every position

---

### ● Menu 1 (default)

|    |            |            |             |
|----|------------|------------|-------------|
| K1 | Mix        | full voice | 100% · 5:00 |
| K2 | Glide      | 0 ms       | 0% · 7:00   |
| K3 | Master vol | unity      | 50% · 12:00 |
| K4 | Vocal vol  | unity      | 50% · 12:00 |
| K5 | Drive      | clean      | 0% · 7:00   |
| K6 | Tone       | flat       | 50% · 12:00 |

---

### ● Menu 2 (hold RIGHT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F1           | 950 Hz  | 62% · 1:15  |
| K2 | F1 bandwidth | 32.5 Hz | 9% · 7:55   |
| K3 | F1 amount    | 1.0     | 50% · 12:00 |
| K4 | F2           | 1800 Hz | 62% · 1:11  |
| K5 | F2 bandwidth | 47.5 Hz | 11% · 8:04  |
| K6 | F2 amount    | 1.0     | 50% · 12:00 |

---

### ● Menu 3 (hold LEFT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F3           | 3350 Hz | 62% · 1:10  |
| K2 | F3 bandwidth | 62.5 Hz | 12% · 8:09  |
| K3 | F3 amount    | 1.0     | 50% · 12:00 |
| K4 | Voices       | 2       | 19% · 8:52  |
| K5 | Detune       | 8 ct    | 13% · 8:20  |
| K6 | Aspiration   | 0.1     | 15% · 8:27  |

---

### ● Toggles

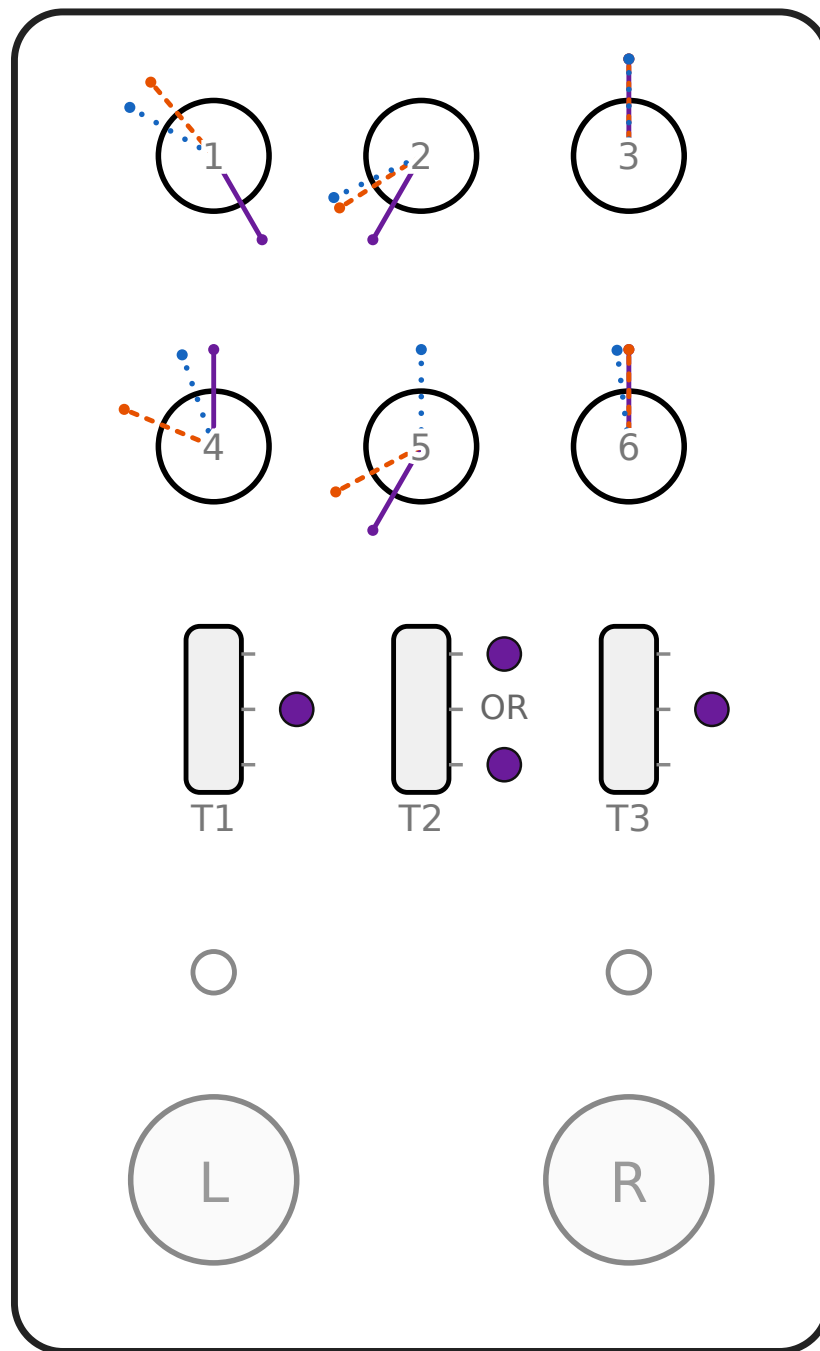
|    |             |                |            |
|----|-------------|----------------|------------|
| T1 | Octave      | 0              | Middle     |
| T2 | Memory page | Set 1 or Set 2 | Up or Down |
| T3 | Gate        | medium         | Middle     |

---

# Master *Old and gravelled*

---

No factory slot: save it to any page



The lowest F1 and F3 of the eight, the widest detune and the most breath of any card. Dark, frayed, faintly ridiculous.

## Master: every position

---

### ● Menu 1 (default)

|    |            |            |             |
|----|------------|------------|-------------|
| K1 | Mix        | full voice | 100% · 5:00 |
| K2 | Glide      | 0 ms       | 0% · 7:00   |
| K3 | Master vol | unity      | 50% · 12:00 |
| K4 | Vocal vol  | unity      | 50% · 12:00 |
| K5 | Drive      | clean      | 0% · 7:00   |
| K6 | Tone       | flat       | 50% · 12:00 |

---

### ● Menu 2 (hold RIGHT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F1           | 640 Hz  | 37% · 10:39 |
| K2 | F1 bandwidth | 32.5 Hz | 9% · 7:55   |
| K3 | F1 amount    | 1.0     | 50% · 12:00 |
| K4 | F2           | 1080 Hz | 28% · 9:45  |
| K5 | F2 bandwidth | 47.5 Hz | 11% · 8:04  |
| K6 | F2 amount    | 1.0     | 50% · 12:00 |

---

### ● Menu 3 (hold LEFT)

|    |              |         |             |
|----|--------------|---------|-------------|
| K1 | F3           | 2400 Hz | 30% · 10:00 |
| K2 | F3 bandwidth | 62.5 Hz | 12% · 8:09  |
| K3 | F3 amount    | 1.0     | 50% · 12:00 |
| K4 | Voices       | 4       | 44% · 11:22 |
| K5 | Detune       | 30 ct   | 50% · 12:00 |
| K6 | Aspiration   | 0.45    | 48% · 11:46 |

---

### ● Toggles

|    |             |                |            |
|----|-------------|----------------|------------|
| T1 | Octave      | 0              | Middle     |
| T2 | Memory page | Set 1 or Set 2 | Up or Down |
| T3 | Gate        | medium         | Middle     |

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## 9 Every gesture

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| To do this                     | Do that                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------|
| Engage or bypass               | Tap either footswitch                                                                              |
| Latch menu 2 (F1, F2)          | Hold RIGHT ~1 s: it latches while you hold                                                         |
| Latch menu 3 (F3, voice stack) | Hold LEFT ~1 s: it latches while you hold                                                          |
| Jump to the other menu         | With one latched, tap the other footswitch                                                         |
| Leave a menu                   | Tap the blinking side's own footswitch                                                             |
| Save to a slot                 | Hold one footswitch past 1 s, then press the other while still holding. The held side is the slot. |
| Charge                         | Engaged, no menu: stomp both together and hold                                                     |
| Firmware update                | <b>Bypassed only:</b> press both together, hold ~2 s                                               |

## TOGETHER, NOT STAGGERED

Both-footswitch gestures need both switches going down at the same time. Holding one for a second and then adding the other is the *save* gesture, and it will save instead.

## What the LEDs mean

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Single colour, both of them: the left LED is always blue, the right always orange. Nothing ever changes an LED's colour, only whether it is off, solid or blinking.

| State              | Left (blue)              | Right (orange) |
|--------------------|--------------------------|----------------|
| Bypassed           | off                      | off            |
| Left slot engaged  | solid                    | off            |
| Right slot engaged | off                      | solid          |
| Freeform engaged   | solid                    | solid          |
| Menu 2 latched     | off                      | blinking       |
| Menu 3 latched     | blinking                 | off            |
| Saved              | 3 blinks                 | 3 blinks       |
| Save refused       | one double-flicker, both |                |
| Charging           | alternating, speeding up |                |

## 10 When it misbehaves

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### **No scream, just my guitar**

Check an LED is solid. Then **knob 1 (Mix)**: at 7:00 you hear only dry signal. Then the **gate (T3)**: on *high* with low-output pickups the voice may never open. Try the middle.

### **A knob does nothing**

Working as designed. After every menu change and every recall the knobs are inert until moved. Nudge it further.

### **It chases the wrong note**

Play one note at a time and mute what you are not using. Open strings ringing under a lead line are the usual culprit. Heavy drive in front makes tracking worse: put DBscreamZ earlier.

### **Both LEDs flickered, nothing saved**

You were on Freeform. It has no slot. Flip T2 to Set 1 or Set 2 and save again.

### **I tried to save and got update mode**

Or the reverse. The gestures differ by timing: save is hold one then add the other; update is both at once, from bypass.

### **It sounds thin and buzzy**

Formant bandwidths set too wide will do that. The cards give the factory values: 7:55, 8:04 and 8:09.

### **The scream lags my playing**

Check **glide** (knob 2). At 5:00 it takes 300 ms to reach each new note. At 7:00 it is instant.

## 11 Specifications

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|            |                                                                              |
|------------|------------------------------------------------------------------------------|
| Platform   | Cleveland Music Co. Hothouse, Daisy Seed3, 125B enclosure                    |
| Processing | 48 kHz, 48-sample blocks, 480 MHz                                            |
| Voice      | FOF formant-grain synthesis, three formants, no noise sources                |
| Tracking   | Cycfi Q bitstream autocorrelation, monophonic                                |
| Bypass     | Buffered, 10 ms crossfade on engage                                          |
| Power      | 9 V DC, 2.1 mm barrel, centre negative                                       |
| Memory     | 4 voice slots plus the global charge configuration, kept across power cycles |

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## Firmware update

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From **bypass**, press both footswitches together and hold about 2 seconds. The pedal appears over USB [universal serial bus] in DFU [Device Firmware Upgrade] mode. Flash it with the Daisy Web Programmer or dfu-util. Press the switches together, not one and then the other.

## Credits

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Formant voice synthesis derived from MonkSynth / Delay Lama. Pitch detection by cycfi/q, Boost Software License 1.0. Hardware from clevelandmusicco/HothouseExamples, open source hardware under CC BY-SA 4.0. The voices are synthesised from measured formant coefficients; no audio is redistributed.

Personal, non-commercial build. The names here are ours and refer to nothing.